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Namespace SmartUIScaler

Classes

CoreUIScaler

Base class of UIScaler components with core scaling logic.

SmartUIAspectFitter

Fits its RectTransform into the parent RectTransform with a fixed aspect ratio or the aspect ratio of its Image or RawImage, anywhere between fitting inside the parent and covering it.

SmartUIAspectSwitch

Switches aspect variants by the aspect ratio of a UIScaler screen area. Listens to the UIScaler on the same object, or the first parent one, and on every resize activates the layout of the matching variant and deactivates the layouts of the other variants. Objects that are not listed as layouts are left untouched. When the active variant changes the deactivated and activated events of the variants are invoked, then the Active Variant Changed event with the name of the new variant, then the listeners added with AddActiveVariantChangedListener(Action<UIScalerAspectVariant>, bool), so any other way of switching can be wired in the inspector or from code. In the editor layouts are switched and the inspector events invoked only in Preview scaling mode, the active variant is resolved in every mode.

SmartUICanvasScaler

UIScaler that fits its RectTransform into the pixel area of the Canvas it is placed under.

SmartUIExpander

Expands its RectTransform from the parent RectTransform to the chosen borders of the Canvas. The whole RectTransform is driven by this component and recalculated on every resize: it covers the parent rectangle, and the chosen edges reach the Canvas borders.

SmartUIFramingBounds

Describes the volume of a character or an object that a SmartUIWorldFollower keeps inside its target rectangle, or that a SmartUIMeshFitter fits into its parent RectTransform instead of measuring the Renderers. The volume is typed in by hand, so it can cover the animations and poses the object will use and does not change while they play. The values are in the space of the nearest SmartUIWorldFollower or SmartUIMeshFitter found on this GameObject or its parents, so framing bounds on equipment held in an animated hand stay still. All enabled framing bounds on that GameObject and its children are combined: equipping an item that has framing bounds of its own widens the framing, and removing it narrows the framing again. The values can be changed at runtime, also by animation clips.

SmartUIManager

Entry point for external code that needs to stay in sync with UIScaler updates. Drives UIScaler updates in play mode and in edit mode. Created automatically and must not be added to a GameObject

manually.

SmartUIMeshFitter

Sizes and places a 3D object, for example a mesh or a whole model, inside the parent RectTransform, so it scales with the UI like any UI element. The bounds of all enabled Renderers on this GameObject and its children are fitted into the parent rectangle after the UIScalers are updated each frame. When SmartUIFramingBounds are found on this GameObject or its children, their typed-in volume is fitted instead, so an animated model keeps the same size while it moves. The back of the bounds is placed on the parent rectangle and moved toward the camera by Depth Offset, measured in UI units, so the depth and the offset scale together with the width and the height. Position and scale of the Transform are set by this component. Rotate a child object instead of this one: rotating this one around X or Y can skew it, because UIScalers do not scale depth. 3D objects are drawn under Screen Space Camera and World Space canvases, not under Screen Space Overlay canvases.

SmartUISizeLimiter

Fits its RectTransform into the parent RectTransform with a fixed reference height and a limited aspect ratio.

SmartUIWorldFollower

Shows a world object inside a RectTransform through a camera of its own, for example a 3D model in a shop slot that lives inside a UIScaler. After the UIScalers are updated each frame the camera projection is shifted so the object's focus point lands in the center of the target rectangle and the camera's field of view (or orthographic size) covers the target height. With Move Camera on, the camera is also placed Distance behind the focus point along its forward direction, so the object is seen head-on. With Move Camera off, the camera transform is left to something else, for example Cinemachine, and only the projection follows the target. With Clip To Target on, the camera viewport is set to the on-screen part of the target, so nothing outside the target is drawn and a target partly off-screen is cut, not squeezed. Turn it off for an overlay camera in a URP camera stack, which ignores its own viewport: the camera then draws its whole viewport and its culling mask must show only the object. The camera belongs to this follower: give it a higher depth than the main camera or add it to the camera stack, and depth only clearing. A target with zero width or height is a point: the object is placed on it and the field of view (or orthographic size) covers the camera viewport height as usual. A camera with a target texture works in that texture's pixels, so the object can be shown in a target inside a Screen Space Camera canvas that renders into the same texture. When SmartUIFramingBounds are found on this GameObject or its children, the projection is also zoomed so their combined volume fits the target, Framing Padding away from its edges, and the center of that volume is used instead of the focus point. The camera's position, rotation and field of view then only decide the viewing angle and the perspective, not the size of the object.

UIScalerBehaviour

Base class of every UIScaler related components.

UIScalerGroupSo
: UIScaler group scriptable object.

UIScalerGroups
: UIScaler groups manager.

UIScalerGroupsPersistenceManager
: Loads the UIScaler group options from PlayerPrefs on Awake and saves them when the application is paused or quit.

UIScalerGroupsSaveData
: UIScaler groups save data.

UIScalerGroupsSaveData.UIScalerGroupOptions
: UIScaler group options save data.

Structs

UIScalerAspectVariant
: Named aspect ratio band of a SmartUIAspectSwitch. The variant with the largest MinAspectRatio that is not wider than the UIScaler screen area becomes active; a screen area thinner than every variant activates the thinnest one. The Layout object is the default way of switching, the events allow any other reaction to be wired in the inspector.

UIScalerOptions
: UIScaler options.

Enums

FramingShape
: Shape of the volume described by SmartUIFramingBounds.

MeshFitMode
: How SmartUIMeshFitter sizes an object inside the parent RectTransform.

SafeAreaSymmetry
: Controls which safe area insets are mirrored to the opposite side of the screen.

UIScalerAlignment
: Used for aligning UIScaler inside provided screen area.

Class CoreUIScaler

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Base class of UIScaler components with core scaling logic.

```
public abstract class CoreUIScaler : UIScalerBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour ← CoreUIScaler

Derived

SmartUICanvasScaler

Inherited Members

UIScalerBehaviour.RectTransform , UIScalerBehaviour.Awake() , UIScalerBehaviour.OnDestroy() , UIScalerBehaviour.OnTransformParentChanged() , UIScalerBehaviour.SetRectTransformDefaults() , UIScalerBehaviour.GetFirstParentCoreUIScaler() , UIScalerBehaviour.WorldRectToLocalRect(Transform, Rect) , UIScalerBehaviour.LocalToWorldPoints(Transform, Vector3, Vector3) , UIScalerBehaviour.WorldToLocalPoints(Transform, Vector3, Vector3) , UIScalerBehaviour.GetWholeScreenRect() , UIScalerBehaviour.GetScreenSafeAreaRect()

Properties

ScreenArea

Screen area the UIScaler was fitted to by its last resize, in screen pixels, after the safe area clamp. Zero until the first resize.

```
public Rect ScreenArea { get; }
```

Property Value

Rect

Methods

AddResizedListener(Action, bool)

Adds a listener invoked after the UIScaler RectTransform was updated by a resize, before the child UIScaler behaviours are resized and before the canvas layout is rebuilt, so objects activated by the listener are fitted in the same frame. The SmartUIAspectSwitch components of this UIScaler have already switched their layouts when it is invoked. A listener that throws is logged and does not stop the other listeners.

```
public void AddResizedListener(Action listener, bool invokeIfResized = true)
```

Parameters

`listener` Action

Listener to add.

`invokeIfResized` bool

Invoke the listener immediately when the UIScaler was resized at least once, so a listener added after the first resize does not wait for the next one.

GetScreenRect()

Called to receive the provided screen area rect.

```
protected abstract Rect GetScreenRect()
```

Returns

Rect

Provided screen area rect.

OnDisable()

```
protected override void OnDisable()
```

OnEnable()

```
protected override void OnEnable()
```

OnPreWillRenderCanvases()

Event that is called just before Canvas rendering happens.

```
protected override void OnPreWillRenderCanvases()
```

RemoveResizedListener(Action)

Removes a listener added with AddResizedListener(Action, bool).

```
public void RemoveResizedListener(Action listener)
```

Parameters

`listener` Action

Listener to remove.

Reset()

```
protected virtual void Reset()
```

Resize()

Method where main resize logic should be implemented.

```
protected override void Resize()
```

ScreenRectToWorldRect(Rect)

Called to convert provided screen area rect into world rect.

```
protected abstract Rect ScreenRectToWorldRect(Rect screenRect)
```

Parameters

`screenRect` Rect

Screen area rect to be converted.

Returns

Rect

World space rect.

Enum FramingShape

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Shape of the volume described by SmartUIFramingBounds.

```
public enum FramingShape
```

Fields

`Box = 1`

Box, which Rotation can turn, for objects that are not upright or not round.

`Cylinder = 0`

Cylinder around the up axis, which Rotation can tilt. Its outline stays the same while the object turns around that axis, so a character on a turntable keeps its size. Tilted, it fits round flat objects such as a shield.

`Sphere = 2`

Sphere. Its outline stays the same however the object turns, so an object that tumbles keeps its size.

Enum MeshFitMode

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

How SmartUIMeshFitter sizes an object inside the parent RectTransform.

```
public enum MeshFitMode
```

Fields

`Envelope = 1`

The object keeps its proportions and covers the parent RectTransform.

`Fit = 0`

The object keeps its proportions and fits inside the parent RectTransform.

`Stretch = 2`

The object is stretched to the width and height of the parent RectTransform.

Enum SafeAreaSymmetry

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Controls which safe area insets are mirrored to the opposite side of the screen.

```
public enum SafeAreaSymmetry
```

Fields

`Both = 3`

Both axes are mirrored.

`Horizontal = 1`

The larger of the left and right insets is applied to both sides.

`None = 0`

The safe area is used as reported, nothing is mirrored.

`Vertical = 2`

The larger of the top and bottom insets is applied to both sides.

Class SmartUIAspectFitter

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Fits its RectTransform into the parent RectTransform with a fixed aspect ratio or the aspect ratio of its Image or RawImage, anywhere between fitting inside the parent and covering it.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUIAspectFitter : UIScalerBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour ← SmartUIAspectFitter

Methods

Awake()

```
protected override void Awake()
```

OnPreWillRenderCanvases()

Event that is called just before Canvas rendering happens.

```
protected override void OnPreWillRenderCanvases()
```

OnTransformParentChanged()

```
protected override void OnTransformParentChanged()
```

Resize()

Method where main resize logic should be implemented.

```
protected override void Resize()
```

Class SmartUIAspectSwitch

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Switches aspect variants by the aspect ratio of a UIScaler screen area. Listens to the UIScaler on the same object, or the first parent one, and on every resize activates the layout of the matching variant and deactivates the layouts of the other variants. Objects that are not listed as layouts are left untouched. When the active variant changes the deactivated and activated events of the variants are invoked, then the Active Variant Changed event with the name of the new variant, then the listeners added with AddActiveVariantChangedListener(Action<UIScalerAspectVariant>, bool), so any other way of switching can be wired in the inspector or from code. In the editor layouts are switched and the inspector events invoked only in Preview scaling mode, the active variant is resolved in every mode.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUIAspectSwitch : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartUIAspectSwitch

Properties

ActiveVariant

Active aspect variant, the default variant with a null Name when there are no variants or the UIScaler was not resized yet. Resolved immediately on enable when the UIScaler was resized before.

```
public UIScalerAspectVariant ActiveVariant { get; }
```

Property Value

UIScalerAspectVariant

ActiveVariantIndex

Index of ActiveVariant in the variants, -1 when none is active.

```
public int ActiveVariantIndex { get; }
```

Property Value

int

ActiveVariantName

Name of ActiveVariant, null when none is active.

```
public string ActiveVariantName { get; }
```

Property Value

string

PreviousVariant

Variant that was active before the last change of ActiveVariant, the default variant with a null Name when none was active before.

```
public UIScalerAspectVariant PreviousVariant { get; }
```

Property Value

UIScalerAspectVariant

VariantCount

Number of aspect variants.

```
public int VariantCount { get; }
```

Property Value

int

Methods

AddActiveVariantChangedListener(Action<UIScalerAspectVariant>, bool)

Adds a listener invoked with the new ActiveVariant when it changes, after the layouts were switched and the variant events were invoked, before the canvas layout is rebuilt. Read PreviousVariant in the listener to know which variant was active before. A listener that throws is logged and does not stop the other listeners.

```
public void AddActiveVariantChangedListener(Action<UIScalerAspectVariant> listener, bool
invokeIfResolved = true)
```

Parameters

`listener` Action<UIScalerAspectVariant>

Listener to add, invoked with the new active variant.

`invokeIfResolved` bool

Invoke the listener immediately when the active variant was resolved at least once, so a listener added after that does not wait for the next change.

GetVariant(int)

Returns the aspect variant at `index`, from 0 to VariantCount - 1, to read its name, aspect ratio or layout.

```
public UIScalerAspectVariant GetVariant(int index)
```

Parameters

`index` int

Returns

RemoveActiveVariantChangedListener(Action<UIScalerAspectVariant>)

Removes a listener added with AddActiveVariantChangedListener(Action<UIScalerAspectVariant>, bool).

```
public void RemoveActiveVariantChangedListener(Action<UIScalerAspectVariant> listener)
```

Parameters

`listener` Action<UIScalerAspectVariant>

Listener to remove.

Class SmartUICanvasScaler

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler that fits its RectTransform into the pixel area of the Canvas it is placed under.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUICanvasScaler : CoreUIScaler
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour ← CoreUIScaler ← SmartUICanvasScaler

Inherited Members

CoreUIScaler.ScreenArea , CoreUIScaler.AddResizedListener(Action, bool) , CoreUIScaler.RemoveResizedListener(Action)

Methods

Awake()

```
protected override void Awake()
```

GetScreenRect()

Called to receive the provided screen area rect.

```
protected override Rect GetScreenRect()
```

Returns

Rect

Provided screen area rect.

OnTransformParentChanged()

```
protected override void OnTransformParentChanged()
```

Resize()

Method where main resize logic should be implemented.

```
protected override void Resize()
```

ScreenRectToWorldRect(Rect)

Called to convert provided screen area rect into world rect.

```
protected override Rect ScreenRectToWorldRect(Rect screenRect)
```

Parameters

`screenRect` Rect

Screen area rect to be converted.

Returns

Rect

World space rect.

Class SmartUIExpander

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Expands its RectTransform from the parent RectTransform to the chosen borders of the Canvas. The whole RectTransform is driven by this component and recalculated on every resize: it covers the parent rectangle, and the chosen edges reach the Canvas borders.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUIExpander : UIScalerBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour ← SmartUIExpander

Methods

Awake()

```
protected override void Awake()
```

OnPreWillRenderCanvases()

Event that is called just before Canvas rendering happens.

```
protected override void OnPreWillRenderCanvases()
```

OnTransformParentChanged()

```
protected override void OnTransformParentChanged()
```

Resize()

Method where main resize logic should be implemented.

```
protected override void Resize()
```

Class SmartUIFramingBounds

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Describes the volume of a character or an object that a SmartUIWorldFollower keeps inside its target rectangle, or that a SmartUIMeshFitter fits into its parent RectTransform instead of measuring the Renderers. The volume is typed in by hand, so it can cover the animations and poses the object will use and does not change while they play. The values are in the space of the nearest SmartUIWorldFollower or SmartUIMeshFitter found on this GameObject or its parents, so framing bounds on equipment held in an animated hand stay still. All enabled framing bounds on that GameObject and its children are combined: equipping an item that has framing bounds of its own widens the framing, and removing it narrows the framing again. The values can be changed at runtime, also by animation clips.

```
[DisallowMultipleComponent]
public sealed class SmartUIFramingBounds : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartUIFramingBounds

Properties

Center

Center of the volume in the space of the SmartUIWorldFollower or SmartUIMeshFitter.

```
public Vector3 Center { get; set; }
```

Property Value

Vector3

Height

Height of the cylinder along its axis in the space of the SmartUIWorldFollower or SmartUIMeshFitter. Used by the Cylinder shape.

```
public float Height { get; set; }
```

Property Value

float

Radius

Radius of the cylinder or the sphere in the space of the SmartUIWorldFollower or SmartUIMeshFitter. Used by the Cylinder and Sphere shapes.

```
public float Radius { get; set; }
```

Property Value

float

Rotation

Rotation of the box or the cylinder in the space of the SmartUIWorldFollower or SmartUIMeshFitter, as Euler angles in degrees, so the volume can lie along a diagonal object such as a sword, or face the camera like a shield. Used by the Box and Cylinder shapes.

```
public Vector3 Rotation { get; set; }
```

Property Value

Vector3

Shape

Shape of the volume.

```
public FramingShape Shape { get; set; }
```

Property Value

FramingShape

Size

Size of the box in the space of the SmartUIWorldFollower or SmartUIMeshFitter. Used by the Box shape.

```
public Vector3 Size { get; set; }
```

Property Value

Vector3

Class SmartUIManager

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Entry point for external code that needs to stay in sync with UIScaler updates. Drives UIScaler updates in play mode and in edit mode. Created automatically and must not be added to a GameObject manually.

```
[InitializeOnLoad]
[AddComponentMenu("")]
[DisallowMultipleComponent]
public sealed class SmartUIManager : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartUIManager

Events

AfterScalersUpdatedEvent

Invoked every frame after UIScalers are updated, and after the SmartUIMeshFitter and SmartUIWorldFollower components have placed their objects and cameras. Read UIScaler sizes and positions here so external UI stays in sync within the same frame.

```
public static event Action AfterScalersUpdatedEvent
```

Event Type

Action

Class SmartUIMeshFitter

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Sizes and places a 3D object, for example a mesh or a whole model, inside the parent RectTransform, so it scales with the UI like any UI element. The bounds of all enabled Renderers on this GameObject and its children are fitted into the parent rectangle after the UIScalers are updated each frame. When SmartUIFramingBounds are found on this GameObject or its children, their typed-in volume is fitted instead, so an animated model keeps the same size while it moves. The back of the bounds is placed on the parent rectangle and moved toward the camera by Depth Offset, measured in UI units, so the depth and the offset scale together with the width and the height. Position and scale of the Transform are set by this component. Rotate a child object instead of this one: rotating this one around X or Y can skew it, because UIScalers do not scale depth. 3D objects are drawn under Screen Space Camera and World Space canvases, not under Screen Space Overlay canvases.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUIMeshFitter : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartUIMeshFitter

Class SmartUISizeLimiter

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Fits its RectTransform into the parent RectTransform with a fixed reference height and a limited aspect ratio.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUISizeLimiter : UIScalerBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour ← SmartUISizeLimiter

Methods

Awake()

```
protected override void Awake()
```

OnTransformParentChanged()

```
protected override void OnTransformParentChanged()
```

Resize()

Method where main resize logic should be implemented.

```
protected override void Resize()
```

Class SmartUIWorldFollower

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Shows a world object inside a RectTransform through a camera of its own, for example a 3D model in a shop slot that lives inside a UIScaler. After the UIScalers are updated each frame the camera projection is shifted so the object's focus point lands in the center of the target rectangle and the camera's field of view (or orthographic size) covers the target height. With Move Camera on, the camera is also placed Distance behind the focus point along its forward direction, so the object is seen head-on. With Move Camera off, the camera transform is left to something else, for example Cinemachine, and only the projection follows the target. With Clip To Target on, the camera viewport is set to the on-screen part of the target, so nothing outside the target is drawn and a target partly off-screen is cut, not squeezed. Turn it off for an overlay camera in a URP camera stack, which ignores its own viewport: the camera then draws its whole viewport and its culling mask must show only the object. The camera belongs to this follower: give it a higher depth than the main camera or add it to the camera stack, and depth only clearing. A target with zero width or height is a point: the object is placed on it and the field of view (or orthographic size) covers the camera viewport height as usual. A camera with a target texture works in that texture's pixels, so the object can be shown in a target inside a Screen Space Camera canvas that renders into the same texture. When SmartUIFramingBounds are found on this GameObject or its children, the projection is also zoomed so their combined volume fits the target, Framing Padding away from its edges, and the center of that volume is used instead of the focus point. The camera's position, rotation and field of view then only decide the viewing angle and the perspective, not the size of the object.

```
[ExecuteAlways]
[DisallowMultipleComponent]
public sealed class SmartUIWorldFollower : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← SmartUIWorldFollower

Enum UIScalerAlignment

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Used for aligning UIScaler inside provided screen area.

```
public enum UIScalerAlignment
```

Fields

`Bottom = 2`

Bottom middle of provided screen area.

`BottomLeft = 7`

Bottom left of provided screen area.

`BottomRight = 8`

Bottom right of provided screen area.

`Center = 0`

Center of provided screen area.

`Left = 3`

Middle left of provided screen area.

`Right = 4`

Middle right of provided screen area.

`Top = 1`

Top middle of provided screen area.

`TopLeft = 5`

Top left of provided screen area.

`TopRight = 6`

Top right of provided screen area.

Struct UIScalerAspectVariant

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Named aspect ratio band of a SmartUIAspectSwitch. The variant with the largest MinAspectRatio that is not wider than the UIScaler screen area becomes active; a screen area thinner than every variant activates the thinnest one. The Layout object is the default way of switching, the events allow any other reaction to be wired in the inspector.

```
[Serializable]
public struct UIScalerAspectVariant : IEquatable<UIScalerAspectVariant>
```

Implements

IEquatable <UIScalerAspectVariant>

Properties

Layout

Object activated while this variant is active and deactivated otherwise, optional.

```
public GameObject Layout { get; }
```

Property Value

GameObject

MinAspectRatio

Smallest aspect ratio of the screen area this variant applies to, as width and height, for example 16:9.

```
public Vector2Int MinAspectRatio { get; }
```

Property Value

Vector2Int

Name

Name reported by ActiveVariantName.

```
public string Name { get; }
```

Property Value

string

Ratio

MinAspectRatio as a single width divided by height value.

```
public float Ratio { get; }
```

Property Value

float

Methods

Equals(UIScalerAspectVariant)

True when `other` has the same Name, MinAspectRatio and Layout; the events are not compared.

```
public bool Equals(UIScalerAspectVariant other)
```

Parameters

`other` UIScalerAspectVariant

Returns

bool

Equals(object)

True when `other` has the same Name, MinAspectRatio and Layout; the events are not compared.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

Returns

bool

GetHashCode()

```
public override int GetHashCode()
```

Returns

int

Class UIScalerBehaviour

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Base class of every UIScaler related components.

```
[RequireComponent(typeof(RectTransform))]
public abstract class UIScalerBehaviour : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerBehaviour

Derived

CoreUIScaler, SmartUIAspectFitter, SmartUIExpander, SmartUISizeLimiter

Fields

RectTransform

RectTransform of this behaviour

```
protected RectTransform RectTransform
```

Field Value

RectTransform

Methods

Awake()

```
protected virtual void Awake()
```

GetFirstParentCoreUIScaler()

Get the first indirect parent CoreUIScaler.

```
protected CoreUIScaler GetFirstParentCoreUIScaler()
```

Returns

CoreUIScaler

Indirect parent CoreUIScaler

GetScreenSafeAreaRect()

Get whole screen safe area.

```
protected static Rect GetScreenSafeAreaRect()
```

Returns

Rect

Whole screen safe area.

GetWholeScreenRect()

Get whole screen area.

```
protected static Rect GetWholeScreenRect()
```

Returns

Rect

Whole screen area.

LocalToWorldPoints(Transform, Vector3, Vector3)

Converts local space points to world space.

```
protected static (Vector3, Vector3) LocalToWorldPoints(Transform localTransform, Vector3
minPoint, Vector3 maxPoint)
```

Parameters

`localTransform` Transform

Local space transform.

`minPoint` Vector3

Bottom left point of rect.

`maxPoint` Vector3

Top right point of rect.

Returns

(Vector3, Vector3)

World space points.

OnDestroy()

```
protected virtual void OnDestroy()
```

OnDisable()

```
protected virtual void OnDisable()
```

OnEnable()

```
protected virtual void OnEnable()
```

OnPreWillRenderCanvases()

Event that is called just before Canvas rendering happens.

```
protected virtual void OnPreWillRenderCanvases()
```

OnTransformParentChanged()

```
protected virtual void OnTransformParentChanged()
```

Resize()

Method where main resize logic should be implemented.

```
protected virtual void Resize()
```

SetRectTransformDefaults()

Sets default values for driven RectTransform.

```
protected void SetRectTransformDefaults()
```

WorldRectToLocalRect(Transform, Rect)

Converts world space rect to local space.

```
protected static Rect WorldRectToLocalRect(Transform localTransform, Rect worldRect)
```

Parameters

`localTransform` Transform

Local space transform.

`worldRect` Rect

World space rect.

Returns

Rect

Local space rect.

WorldToLocalPoints(Transform, Vector3, Vector3)

Converts world space points to local space.

```
protected static (Vector3, Vector3) WorldToLocalPoints(Transform localTransform, Vector3
minPoint, Vector3 maxPoint)
```

Parameters

`localTransform` Transform

Local space transform.

`minPoint` Vector3

Bottom left point of rect.

`maxPoint` Vector3

Top right point of rect.

Returns

(Vector3, Vector3)

Local space points.

Class UIScalerGroupSo

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler group scriptable object.

```
[CreateAssetMenu(fileName = "UIScalerGroup", menuName = "SmartUIScaler/UIScalerGroup")]
public class UIScalerGroupSo : ScriptableObject
```

Inheritance

object ← Object ← ScriptableObject ← UIScalerGroupSo

Properties

DefaultOptions

Default options of this group.

```
public UIScalerOptions DefaultOptions { get; }
```

Property Value

UIScalerOptions

SaveId

Identifier used to match this group in save data. Generated when the asset is created and editable in the inspector. The editor keeps it unique across the project. A group with an empty Save Id is not persisted.

```
public string SaveId { get; }
```

Property Value

string

Methods

SetContentScale(float)

Sets the content scale for this group.

```
public void SetContentScale(float contentScale)
```

Parameters

`contentScale` float

Controls the scale of the objects inside UIScalers. Recommended scale is [0.1, 1.0].

SetExpandProgress(float)

Sets the expand progress for this group.

```
public void SetExpandProgress(float expandProgress)
```

Parameters

`expandProgress` float

Controls UIScaler occupied space between reference resolution and max allowed screen space. Allowed value is [0.0, 1.0].

Class UIScalerGroups

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler groups manager.

```
public static class UIScalerGroups
```

Inheritance

object ← UIScalerGroups

Fields

PlayerPrefsSaveKey

Default PlayerPrefs key for saving UIScaler groups save data.

```
public const string PlayerPrefsSaveKey = "UIScalerGroupsSaveData"
```

Field Value

string

Methods

AddGroupListener(UIScalerGroupSo, Action<UIScalerOptions>, bool)

Add listener to UIScaler group options changes.

```
public static UIScalerOptions AddGroupListener(UIScalerGroupSo uiScalerGroup,
Action<UIScalerOptions> listener, bool invokeListenerOnAdd = true)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to listen to.

`listener` Action<UIScalerOptions>

Changes callback.

`invokeListenerOnAdd` bool

If change callback should be invoked on add.

Returns

UIScalerOptions

Current UIScaler group options.

GetSaveData()

Get current UIScaler groups save data. Use it for custom saving. Includes loaded options of groups that have not been used yet, so they are not lost. Groups with an empty Save Id are skipped. When several groups share a Save Id only the first used one is saved.

```
public static UIScalerGroupsSaveData GetSaveData()
```

Returns

UIScalerGroupsSaveData

UIScaler groups save data.

LoadFromPlayerPrefs(string)

Load UIScaler groups save data from PlayerPrefs to be applied to UIScalers.

```
public static void LoadFromPlayerPrefs(string playerPrefsSaveKey = "UIScalerGroupsSaveData")
```

Parameters

`playerPrefsSaveKey` string

PlayPrefs key.

RemoveGroupListener(UIScalerGroupSo, Action<UIScalerOptions>)

Remove UIScaler group options changes listener.

```
public static void RemoveGroupListener(UIScalerGroupSo uiScalerGroup,
Action<UIScalerOptions> listener)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to remove listener from.

`listener` Action<UIScalerOptions>

Changes callback.

SaveToPlayerPrefs(string)

Save current UIScaler groups save data to PlayerPrefs.

```
public static void SaveToPlayerPrefs(string playerPrefsSaveKey = "UIScalerGroupsSaveData")
```

Parameters

`playerPrefsSaveKey` string

PlayPrefs key.

SetGroupContentScale(UIScalerGroupSo, float)

Set UIScaler group content scale.

```
public static void SetGroupContentScale(UIScalerGroupSo uiScalerGroup, float contentScale)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to apply content scale.

`contentScale` float

Controls the scale of the objects inside UIScalers. Recommended scale is [0.1, 1.0].

SetGroupExpandProgress(UIScalerGroupSo, float)

Set UIScaler group expand progress.

```
public static void SetGroupExpandProgress(UIScalerGroupSo uiScalerGroup,
float expandProgress)
```

Parameters

`uiScalerGroup` UIScalerGroupSo

UIScaler group to apply expand progress.

`expandProgress` float

Controls UIScaler occupied space between reference resolution and max allowed screen space. Allowed value is [0.0, 1.0].

SetSaveData(UIScalerGroupsSaveData)

Set UIScaler group save data to be applied to UIScalers. Use it for custom loading. Options of groups that are not used yet are applied when they are first used. The provided save data is not modified.

```
public static void SetSaveData(UIScalerGroupsSaveData saveData)
```

Parameters

`saveData` UIScalerGroupsSaveData

UIScaler groups save data.

StoreSaveDataToPlayerPrefs(string)

Store current UIScaler groups save data to PlayerPrefs without saving.

```
public static void StoreSaveDataToPlayerPrefs(string playerPrefsSaveKey
= "UIScalerGroupsSaveData")
```

Parameters

`playerPrefsSaveKey` string

PlayPrefs key.

Class UIScalerGroupsPersistenceManager

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

Loads the UIScaler group options from PlayerPrefs on Awake and saves them when the application is paused or quit.

```
public sealed class UIScalerGroupsPersistenceManager : MonoBehaviour
```

Inheritance

object ← Object ← Component ← Behaviour ← MonoBehaviour ← UIScalerGroupsPersistenceManager

Class UIScalerGroupsSaveData

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler groups save data.

```
[Serializable]
public class UIScalerGroupsSaveData
```

Inheritance

object ← UIScalerGroupsSaveData

Fields

CurrentVersion

Format version written by GetSaveData(). Data written by plugin version 1.0 has no version and no Save Id values and is ignored on load.

```
public const int CurrentVersion = 2
```

Field Value

int

GroupOptions

List of UIScaler group options save data.

```
public List<UIScalerGroupsSaveData.UIScalerGroupOptions> GroupOptions
```

Field Value

List <UIScalerGroupsSaveData.UIScalerGroupOptions>

Version

Format version of this data. See CurrentVersion.

```
public int Version
```

Field Value

int

Class UIScalerGroupsSaveData.UIScalerGroupOptions

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler group options save data.

```
[Serializable]
public class UIScalerGroupsSaveData.UIScalerGroupOptions
```

Inheritance

object ← UIScalerGroupsSaveData.UIScalerGroupOptions

Fields

ContentScale

Controls the scale of the objects inside UIScalers.

```
public float ContentScale
```

Field Value

float

ExpandProgress

Controls UIScaler occupied space between reference resolution and max allowed screen space.

```
public float ExpandProgress
```

Field Value

float

SaveId

Save identifier of UIScalerGroupSo scriptable object. See SaveId.

```
public string SaveId
```

Field Value

string

Struct UIScalerOptions

Namespace: SmartUIScaler

Assembly: SmartUIScaler.dll

UIScaler options.

```
[Serializable]
public struct UIScalerOptions : IEquatable<UIScalerOptions>
```

Implements

IEquatable <UIScalerOptions>

Fields

Alignment

Controls the alignment of UIScaler inside provided screen area.

```
public UIScalerAlignment Alignment
```

Field Value

UIScalerAlignment

ContentScale

Controls the scale of the objects inside UIScalers. Recommended scale is [0.1, 1.0].

```
public float ContentScale
```

Field Value

float

ExpandProgress

Controls UIScaler occupied space between reference resolution and max allowed screen space. Allowed value is [0.0, 1.0].

```
[Range(0, 1)]
public float ExpandProgress
```

Field Value

float

ReferenceResolution

The resolution the UI layout is designed for.

```
public Vector2Int ReferenceResolution
```

Field Value

Vector2Int

SafeAreaSymmetry

Mirrors the larger safe area inset of the chosen axes to the opposite side, so the UIScaler stays centered on the screen and split screen UIScalers keep the same scale when the safe area is one-sided. Used only when UseSafeArea is enabled.

```
public SafeAreaSymmetry SafeAreaSymmetry
```

Field Value

SafeAreaSymmetry

UseSafeArea

Should UIScaler use only the screens safe area or whole provided screen area.

```
public bool UseSafeArea
```

Field Value

bool

Properties

Default

Default UIScaler options.

```
public static UIScalerOptions Default { get; }
```

Property Value

UIScalerOptions

Methods

Equals(UIScalerOptions)

Checks if all options are equal to the options of another instance.

```
public bool Equals(UIScalerOptions other)
```

Parameters

`other` UIScalerOptions

Options to compare with.

Returns

bool

True when every option is equal.

Equals(object)

Checks if the object is UIScaler options with all options equal to this instance.

```
public override bool Equals(object obj)
```

Parameters

`obj` object

Object to compare with.

Returns

bool

True when the object is equal UIScaler options.

GetHashCode()

Hash code combined from all options.

```
public override int GetHashCode()
```

Returns

int

Hash code.